Operational Diagnosis, Single Source of Truth & AI Reconciliation Strategy

Candidate Submission | Role: AI Operations Associate | Date: 30 June 2026 Snapshot | Constraint: Zero New Headcount

Repo: https://github.com/Rudra117/bharattrip-refund-ops | Prototype Run: python -m uvicorn api:app --port 8000 (or streamlit run app.py)

CORE OPERATIONAL ASSUMPTIONS (DOCUMENTED UPFRONT):

1. ID Normalization: Strip prefixes/spaces, 4-digit pad (RF-0421 -> RF-0421). Deduplication preserves latest update, then Finance match.
2. Approval Tier: Inferred at >= INR 20,000 (all approved payouts >= 20,100). 3. Ground Truth: Deductions explain gross-to-net variances.

1. DEFINE THE PROBLEM: DATA OBSERVATIONS vs. REPORTED SYMPTOM

Leadership's stated symptom is that 'escalations have been climbing for two months while refund volume is flat.' The underlying diagnosis is an operational disconnection between two independent spreadsheet trackers rather than external volume growth.

Refund request demand remained completely flat across all 5 months (~145/month: Feb 144, Mar 152, Apr 147, May 148, Jun 142). The escalation surge in May and June is driven by three distinct internal systemic failures:

1. Gap Class B (Closed in Support, No Finance Record): Support agents marked tickets 'Closed' with notes like 'refund processed, informed agent'. However, no corresponding record exists in Finance Tracker. This rate increased from 2.1% in March to 18.7% in May and 33.0% in June. Across the snapshot, exactly 69 closed tickets with no Finance record exist, totaling INR 1,080,700.
2. Finance Processing Turnaround & Approval Bottleneck: Turnaround on completed payouts increased from 2.67 days in April to 6.52 days in May (unpaid refunds are excluded from turnaround calculations). In June, an approver was recorded on almost no payouts (1.7%; 1 payout) because refunds above ~INR 20,000 (the inferred approval tier) were mostly still pending; cause not visible in the data.
3. Recorded Deductions on Disbursements: Finance recorded Deductions (averaging INR 1,500-2,600). Support records do not document these deductions, leading travel agents to escalate regarding differences between requested and credited amounts.

2. OPERATIONAL ANALYSIS: EMPIRICAL METRICS ACROSS 5 MONTHS

Reconciling 733 unique Support rows and 680 Finance rows reveals 749 unique refund cases. While incoming volume stayed steady, the operational handoff broke down in May and June:

Month	Requests	Support Clsd	Gap B (Count)	Gap B Rate	Avg Proc Time*	Finance Done	Approver Share**
Feb 2026	144	137	4	2.9%	2.89 days	111	32 (28.8%)
Mar 2026	152	146	3	2.1%	2.87 days	154	46 (29.9%)
Apr 2026	147	145	6	4.1%	2.67 days	139	41 (29.5%)
May 2026	148	139	26	18.7%	6.52 days	120	29 (24.2%)
Jun 2026	142	91	30	33.0%	4.88 days	59	1 (1.7%)

**Unpaid refunds excluded from turnaround calculations. **Approver share denominator = all Finance Refund Done rows by received month.*

Key Takeaway from Table:

In June, completed payouts dropped to 59, while Support closed tickets with no Finance record surged to 30 (33.0% of Support closures). Support agents were telling agents refunds were completed, but Finance never received the tickets, sparking intense follow-up chases.

2. OPERATIONAL ANALYSIS (CONT.): WHERE TRACKERS AGREE, DISAGREE, OR FALL SILENT

- Where Trackers Agree (664 cases): Captured in both systems; 571 completed payouts recorded in Support.
- Where Trackers Disagree (18 cases): 12 tickets are Closed in Support but Pending Payout in Finance (Gap C: INR 186,000); 6 tickets are Closed in Support but Declined in Finance (Gap D: INR 94,800). Additionally, 50 paid cases exhibit amount differences (Gap E); recorded deductions explain 100% of these 50 variances (Support Amount minus Finance Paid equals recorded Deduction).
- Where Trackers Fall Silent: Support falls silent on Finance in 69 cases (Gap B: INR 1,080,700). Finance falls silent on Support in 16 paid refunds (Gap H: INR 141,800). 25 high-value refunds (>= INR 20,000) totaling INR 696,600 sit stalled in Finance approval (mean wait 14.2 days).

INFORMAL OFF-TRACKER REQUESTS FROM THE BRIEF (REAL ROOT CAUSE CONFIRMATION):

1. Peak Journeys (Email, 5 June): DEL-DXB cancelled, no reference number issued. Chased repeatedly until logged late as RF-1099.
2. Nomad Travel (WhatsApp, 8 June): 'Bhai refund ka kya hua? BLR-MAA wala. 2 hafte ho gaye'. Off-tracker chase, later logged as RF-1098.
3. Support Agent Note (Email, 11 June): 'Finance says they never received several refunds I closed last month. I marked them done on my side. Not sure where they went.' Direct confirmation that Support closed tickets without ensuring Finance ingestion.

Root Cause Breakdown of All 155 Logged Escalations:

Root Cause Category	Count	Share (%)	Data Observation & Action
paid and matched	57	36.8%	Payout completed matching request; agent inquired regarding confirmation / missing UTR.
Support closed/no Finance	31	20.0%	Gap Class B: Ticket marked Closed on Support, no corresponding Finance record.
never logged (no usable ref)	29	18.7%	Informal inquiries (email/WhatsApp) with no usable reference in operational trackers.
paid but amount differs	18	11.6%	Disbursement matches Amount Paid, but agent expected gross amount without deduction.
Finance pending/Support not closed	8	5.2%	Request in Finance processing queue while Support ticket remains active.
declined (recorded in Finance)	6	3.9%	Decline recorded in Finance; decline details not documented on Support side.
Support closed/Finance pending	6	3.9%	Premature closure: Support marked Closed while Finance record is Pending Payout.

Estimated Addressable Failures: 141 of 155 (91.0%) = 84 structural failures (31 Gap B + 29 unlogged + 18 deductions + 6 premature closures) + 57 status queries on p

Validation Wishlist (What to validate with 2 weeks more access):

1. Finance Approval Workflow: Review approval logs to verify the inferred INR 20k threshold and identify why large June payouts stalled.
2. Deduction Schedule Table: Ingest standardized cancellation fee tables to verify deductions prior to payout submission.
3. Informal Request Inboxes: Access incoming email and WhatsApp logs to measure total volume of unlogged agent inquiries.
4. Canonical Agent Master: Unify agent naming variations across systems into a single entity registry.

3. DESIGN A SOLUTION: PROCESS & SINGLE SOURCE OF TRUTH (ZERO HEADCOUNT)

Under the zero-new-headcount constraint, BharatTrip must eliminate redundant manual status updates rather than hire coordinators:
- Single Unified Table: Support and Finance retain role-specific views, but read and write to a single reconciled database backend. Support status is restricted to 'New', 'Working', and 'Verified & Forwarded'. 'Closed' is a system-generated state triggered strictly when Finance enters payout execution ('Refund Done') with bank UTR.
- Automated Reconciler & Action Queue (core/reconcile.py): Daily reconciliation identifies discrepancies, automatically routing the 69 Gap B records to Finance and ranking open tasks by Priority Score (Amount x Age).
- Proactive Dispatch Notification: On payout execution, automated notification sends the agent the bank transaction reference and an itemized breakdown (Gross Amount, Deduction, Net Disbursed), addressing the pattern behind 57 status queries and 18 deduction disputes.

Component	Operational Choice	Justification & Risk Control
Matching & Reconciliation	100% Deterministic Code (NO LLM)	Zero financial hallucination risk. Ledger math and status joins remain deterministic.
Deduction Verification	Deterministic Rule Engine	Validates whether Refund Amount minus Amount Paid equals Deduction exactly.
Informal Intake (WhatsApp/Email)	AI / LLM Layer (core/intake.py)	Normalizes unstructured English/Hinglish text into structured JSON schema.
Agent Status Response	Automated Grounded Drafts	Strictly grounded in ledger records. Never invents an estimated turnaround (ETA).
Disbursement Authorization	Deliberately Kept Manual (Finance)	Banking authorization and payment release remain with authorized Finance personn
Decline Communication	Manual Support Touchpoint	Human touchpoint preserves partner agency relationships on contested cancellation

AI Prototype Implementation (Judgment & Functional Scope):

We built a functional production platform (FastAPI + React/Tailwind + Streamlit fallback) backed by modular deterministic code and Gemini AI. It solves the exact unlogged intake and reconciliation slice: parses unstructured Hinglish/English emails and WhatsApp messages, queries the reconciled ledger, runs plausibility checks (prevents asserting false status on aged records like Preset b), routes internal notes to Gap B reports, and auto-assigns non-colliding RF-NEW-### IDs. Built-in zero-key fallback guarantees 100% operational continuity without API keys.

5. MEASURE: 30-DAY SUCCESS METRICS & TARGET RESULTS

Primary Metric to Watch	June Baseline	30-Day Target	Operational Significance
1. Gap B Creation Rate	33.0% (30 tickets)	< 1.0% (< 2 tickets)	Direct measure of handoff synchronization between Support and Finance.
2. Escalation Frequency	35 per 100 refunds	< 5 per 100 refunds	Proactive deduction breakdowns eliminate repeated status queries.
3. Backlog Aging (>14 days)	69 tickets > 30 days	0 tickets > 14 days	Action Queue prioritizes aged high-value items.
4. Intake Auto-Logging Share	~0% (all manual)	> 95% logged	All off-tracker email/WhatsApp requests logged with RF-NEW-###.
5. Stalled Approvals (>7 days)	25 tickets (INR 6.96L)	0 tickets > 7 days	Approval alerts prevent month-end large refund bottlenecks.

Target Result After 1 Month:

- Discrepancy Elimination: Gap B creation drops to zero via system-enforced state machine (Support cannot close without Finance UTR).
- Backlog Liquidation: The 69 Gap Class B refunds (INR 1,080,700) and 25 stalled approvals (INR 696,600) audited and disbursed.
- Escalation Reduction: Escalations drop from ~35 per 100 refunds to under 5 per 100 refunds as agents receive real-time UTR and fee breakdowns.